

The Trezo Engine

Agents, gates, formulas, strategies — and what it cannot yet do

This revision continues the reference of 23 August 2026 (commit eb271bb). The sixteen days between the two editions were the loudest in the platform's history: a four-day silent approval outage, a phantom-close loop that booked losses that never happened, the first working alert channel, two silent-failure nets that have since fired correctly in production, per-book reward:risk floors, real-venue crypto with resting stops, and a capacity lock that revealed how full the books really are. Every rule below is read from the deployed tree at its real value; Part 8 is the record of what changed and why, and Part 7 still lists what is missing, modelled or inert — because a reference that only describes the intended behaviour is the kind of document that lets a silent failure survive for months.

Revision commit 4163078 · 8 September 2026 · supersedes eb271bb (23 August 2026)

Scope the agents service — scanners, risk, execution, position management, operations

Source read from the deployed tree, not from design notes

Prepared by Nova, for Mike

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Eight parts. The first three describe the machine; the next two describe what it knows; six and seven describe what it trades and where it is weak; the eighth is new — the record of the sixteen days since the first edition.

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HOW TO READ A NUMBER IN THIS DOCUMENT

Every threshold quoted here was read from the source at commit 4163078 and is written at its deployed value, not its documented intent. Where the two disagree, the disagreement is called out. Constants appear in monospace. A box like this one in amber is an uncomfortable truth; red is a live defect; green is a defect from the first edition that is now closed, kept so the closure is checkable.

1 How the engine is put together

A signal is born in a scanner, judged once by a central gate, routed to a broker by a single executor, and then owned by a monitor until it closes. Nothing skips a step — and since this revision, nothing dies quietly either.

1.1 The loop

Every trade travels the same four stages, and the separation is deliberate: the component that *finds* an idea is never the component that *approves* it, and neither one places the order.

SCANNERS (11 agents) propose signals	->	RISK MANAGER (single gate) approve / veto size the trade	->	TRADE EXECUTION (single router) route to broker PER BOOK	->	POSITION MONITOR (60-second tick) stops, targets, trims, reconcile
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What changed since eb271bb: the executor now fans an approval out per book and answers for every one of them. A book that already holds the name, a book at its position cap, and a pocket at its slot cap each refuse *audibly* — an activity row naming the book, the pocket and the open/cap numbers — because on 3 September eight stock approvals died in silence and cost an hour of live diagnosis on a lane that was behaving perfectly.

1.2 Books and pockets

Capital is divided twice. The **book** is an independent account with its own equity, settings and open positions; the **pocket** is a market-type budget inside a book. Three books run today: primary (~\$5k), 25k, and 75k, all paper.

POCKET BUDGETS BY POSTURE — WIDENED 2 SEPTEMBER (MIKE): THE STOCK POOLS GREW IN EVERY POSTURE

POSTURE	CRYPTO	STOCK	OPTION	INCOME	FOREX	AUTO-SELECTED WHEN
growth	28%	50%	9%	8%	5%	equity < \$25,000
balanced	15%	45%	16%	20%	4%	\$25,000 – \$100,000
income	8%	28%	18%	42%	4%	equity ≥ \$100,000
velocity	36%	30%	20%	7%	7%	never — opt-in only (24/7 + fast-settling lanes)

A pocket is a hard dollar budget: $\text{budget} = \text{equity} \times \text{split}$; a trade whose notional would exceed what remains is skipped, not shrunk. Below \$25,000 the budgets soften — stretched 1.75× and capped at 60% of equity per lane. Position slots divide the same way: a book's `max_open_positions` (default now 14, up from 3 in the first edition) is shared out in proportion to pocket size; an unfunded pocket gets zero slots, every funded pocket keeps at least one.

SEPTEMBER'S LESSON: A FULL BOOK IS A STATE, NOT A FAULT

Over Labor Day weekend every book's crypto pocket sat at or over cap (3/2 and 6/6) and the stock pockets too (5/4, 6/5, 3/3), so every approval was refused — correctly. The flow alarm read those deliberate refusals as "no outcome at all" and pinged urgent every half hour. The fix (on main, awaiting deploy): a deliberate refusal is an outcome, and a window whose approvals were ALL refused reports once as a warn CAPACITY LOCK — free a slot or raise a cap — never as a malfunction.

1.3 The bus and the registry

Agents do not call each other. Every message goes onto a bus that fans it to every other enabled agent's `on_message`. The registry holds one entry per agent — enabled flag, last tick, tick count, last error, actor or observer. Scheduling is `APScheduler` with the same three settings that matter (`max_instances=1`, `coalesce=True`, `misfire_grace_time=None`) — plus, since 29 August, **per-agent tick ceilings derived from observed cancellations**: the old flat 900s ceiling was silently cancelling every options-scanner tick, which is how the wheel went quiet for weeks without an error.

1.4 Fail-open is a design rule — and now it is audible

The advisory components still all **allow** on exception, missing data or disabled flag: a new advisory component must never be able to freeze a lane that worked before it existed. What changed is the cost. Two nets now make the silent path loud:

Net 1 — a handler exception is announced. The bus router publishes `handler_failed` (as the failing agent), writes an activity row, and pings the webhook once per (agent, error). **Net 2 — the flow itself is counted, per lane.** Signals in with zero approvals out, or approvals in with zero fills out, during that lane's market hours, is an alarm on its own, whatever the cause. Both nets have fired correctly in production: Net 2 caught a real 26-minute crypto approval-starvation window on 2 September and the Discord alert delivered.

WHY THE NETS EXIST

From 27 to 31 August the risk manager raised `UnboundLocalError` on every signal carrying a real direction. The router swallowed it to stdout. No bus message, no activity row, no alert — the platform approved NOTHING for four trading days while every health check said every agent was ticking. The log looked quiet, not broken. The tell, visible four days early, was `trade_execution's msgs=0` in `report_status`: when a lane goes silent, count APPROVES.

1.5 Paper and live

Real money still has a single chokepoint: `_LIVE_EXECUTOR_AVAILABLE = False` is a module constant, so `live_trading_enabled()` is always false in this build and setting `TRADING_MODE=live` changes nothing real. Unchanged.

CRYPTO IS NO LONGER MODELLED — IT TRADES AT THE REAL VENUE

The first edition said crypto was fully modelled because Kraken had no paper API. Since then the lane routes to Alpaca's paper crypto venue (`alpaca_crypto_enabled`, with the internal modelled engine kept as the automatic fallback if the connector errors). This is a real API with real market data, real order semantics, and real rejections — resting stop_limit GTC stops ratcheted by the monitor, 403s for insufficient balance when a quantity is rounded a hair past the holding, and 403 wash-trade rejections when a new order meets a resting stop. The lane's P&L is no longer the platform grading its own homework.

1.6 The operational spine (new since eb271bb)

The ops relay is a Supabase mailbox the engine drains every five minutes — exactly six CHECK-constrained job kinds (`git_pull_restart`, `pip_install`, `report_status`, `restart_service`, `tail_log`, `web_rebuild`). It is not a shell and cannot trade. The reverse channel pushes the activity log to Supabase so it is readable from anywhere. **The beacon rule:** a

deploy is done when a NEW process says hello — engine_boot names pid, commit and agent count; a job row saying "done" only means the pull happened. **The deploy gate** runs every guard suite in one bare process (no pytest, no .env, no network) and a red gate rolls the checkout back automatically. **The alert channel** (TREZO_ALERT_WEBHOOK, Discord/Slack auto-detected) is the exit the platform never had: severity-tiered dedupe — urgent repeats every 30 min while the condition holds, warn every 4h, info daily — and every send or failure also writes an activity row so the local record is complete when the channel is down.

2 The agent roster

Thirty registered agents — market_desk joined since the first edition. Five may place orders; the rest propose, advise or observe. The distinction is enforced at registration, and the watchdog's expected roster now matches the registry exactly, so no agent can go quiet without something noticing.

GROUP	AGENTS	NOTES
Scanners — propose, never act	pattern_detection (180s), stms_scanner (180s), orb_scanner (120s), extended_scanner (1800s), crypto_scanner (180s), forex_scanner (180s), options_scanner (1800s), dividend_lt (1800s)	Each publishes ticker + direction + strategy + TCS + proposed geometry.
The spine — judge, route, manage	risk_manager (event), trade_execution (event), position_monitor (60s)	The gate, the router, the owner of everything open. Event-driven agents never tick on a timer.
Order authority besides the spine	options_scanner (wheel), dividend_manager (21600s), kindrip (modelled)	Five actors total: options_scanner, trade_execution, position_monitor, kindrip, dividend_manager.
Context — change what the spine believes	adaptive_scope (600s), market_sentiment (1800s), market_horizon (900s), cycle_awareness (21600s), research (1800s), relay_ingest (300s), market_desk (NEW)	market_desk reads the scheduled market reports (08:00 / 13:00 / 15:00 ET) so every lane gets a view of the tape.
Review — measure and report	portfolio_architect (21600s), strategy_discovery (3600s), exit_advisor (300s), exit_advisor_options (300s), tax_optimizer (1800s), broker_truth (900s), book_health (300s)	Read-only. broker_truth reconciles what Alpaca actually holds.
Operations	ops_watchdog (300s), archivist (900s), user_support (event)	Watchdog: registry audit, per-agent silence, flow nets, relay drain, log push. user_support remains a registered stub (INERT).

THE SIX-AGENT MONITORING GAP IS CLOSED

The first edition flagged that the watchdog's hand-maintained expected roster held 23 names against 29 registered, so six agents could go silent unnoticed — which is exactly how two books ran for six days with no watchlist. The expected roster now carries all thirty and is asserted against the registry by a guard suite.

Load-bearing non-agents, unchanged: **activity_log** (fail-open JSONL ledger of every gate decision — and now of every deliberate refusal), **reevaluator** (position re-scoring; its first-sight heartbeat no longer waits out the machine's first hour after boot), **base** (the Agent class and the message type). The crypto re-scoring half of the 2 September defense ships behind TREZO_CRYPTOREEVAL, still off in production as of this revision — the MAE ceiling and time limits apply, the re-score is dormant.

3 The gates

Roughly forty rules can stop a trade. A gate is not a preference: most exist because something went wrong once, and the number is the scar. Where the history is known it is stated, because a threshold without its reason is a threshold somebody will "optimise" later.

3.1 Capital and concentration

RULE	VALUE	PROTECTS AGAINST
A book will not open a new ticker past its own cap	<code>max_open_positions</code> default 14	One book's count judging another's — the failure that killed 516 entries in a day. (Default was 3 in the first edition; raised with the posture split.)
Each pocket gets its proportional share of slots	<code>max(1, round(cap × alloc / total))</code>	A hot lane occupying another lane's chairs
Each market type has a hard dollar budget	<code>equity × POSTURE_SPLIT</code> (\$1.2, widened 9/2)	Any single lane eating the whole account
Per-position notional cap, scaled by account size	<code><\$10k 15% · <\$25k 30% · <\$100k 25% · ≥\$100k 15%</code>	Low-volatility names with tiny stop distances taking ~58% of capital in one order
Total deployment ceiling	<code>1.25 × equity</code>	Unbounded margin leverage
A single crypto coin cannot exceed a share of equity	<code>10%, summed across all open rows</code>	Cross-day accumulation concentrating the book in one coin
Crypto accumulation adds must be spaced	<code>18 hours</code>	"Buy the dip" firing every scanner tick
The same ticker cannot be approved twice while open	<code>TTL 2 hours — PER BOOK since 2 Sep (981c00f)</code>	One name approved three times in a morning. The dedup was platform-wide, so one book's approval silenced the same idea for every other book.
Lottery-ticket option structures capped	<code>3% of options capital</code>	Allocation creep into low-probability plays
Below the hard-pocket floor, budgets stretch then cap	<code>×1.75, max 60% of equity</code>	A pocket too small to buy one position at all

3.2 Liquidity and tradability

RULE	VALUE	PROTECTS AGAINST
Average daily volume floor, per strategy	<code>STMS 100k · ORB 500k · Extended 500k · Wheel 500k · global default 250k</code>	Unfillable names
Bid/ask spread wider than the limit blocks the trade	<code>0.5%</code>	Slippage on illiquid names
No live quote and no recent trade reads as a halt	<code>tape stale > 20 min</code>	Trading into a halt

RULE	VALUE	PROTECTS AGAINST
Last bar far from quoted mid = spread unmeasured, not wide	2% divergence	The "10.62% spread" artifact — 166 false vetoes in one day
Scalps reserved for liquid capitalisation tiers	micro and unknown excluded	Scalping a name whose spread eats the edge
Small-cap momentum screen	\$1-\$20 · +10% move · 5× rel. volume · float < 20M	Wrong-shaped names entering the momentum lane
Dividend quality screen	payout ≤ 70% · yield ≥ 1.5% · fund volume ≥ \$2M/day	Dividend traps and untradeable funds

THE STALENESS TIERING NOW FIRES (WAS: UNREACHABLE)

The first edition found the staleness bands still testing $tcs \geq 700$ and ≥ 500 on a 0–100 score, so every signal fell through to a flat 300 seconds. The bands are rescaled: $TCS \geq 70$ expires in 60s, 50–69 in 180s, below 50 in 300s.

3.3 Price extension and timing

RULE	VALUE	PROTECTS AGAINST
Will not chase a stock too far from its 20-day average	$4.0 \times ATR(14)$	Buying the top of a parabolic move
Longs blocked when SPY and QQQ both below session VWAP; shorts when both above	cache 120s	Fighting the broad tape
Targets capped at what the name actually moves, then floored	$1.5 \times ATR\%$, floor 0.6%	"Waiting money" — a 10% target on a name that moves 2%
Intraday time stop, and stagnation stop	90 min hold · 75 min at <0.25R	Dead money occupying a slot. Crypto exempt.
End-of-day force exit for intraday strategies	19:45 UTC (3:45 PM EDT – drifts an hour under EST; see 7.2)	Overnight gap risk on a day trade
Multi-day swing time stop	7 days	Swings becoming accidental investments
Adopted-row shield	TREZO_ADOPTED_TIME_EXIT default off	A row adopted with an old true entry being market-sold on its first tick by an honest clock (see 4.13)

3.4 Risk per trade and portfolio limits

RULE	VALUE	PROTECTS AGAINST
Kill switch — daily realised loss	3% of day-start equity	Bleeding out in a single session
Kill switch — weekly realised loss	6% of week-start equity	A bad week compounding

RULE	VALUE	PROTECTS AGAINST
Kill switch — consecutive losses, per book	3	Tilt, or a broken thesis repeating
Kill switch — broker rejects, rolling window	3 in 60 min	A reject storm
Kill switch — execution quality	3 fills worse than 75 bps	A market that will not give your price
Risk per trade	Bot Tuning slider, clamp 0.05%-25%; absent slider falls back to 1%	Outsized single-trade loss
Reward-to-risk floor — PER BOOK since 2 Sep	book's min_reward_risk; seed 1.5; clamp 0.3-3.0; reject below floor - 0.011	Taking setups that need an impossible win rate. Books run 0.4 (primary) and 0.5 (25k/75k) today.
Reward:risk harmoniser	stop = target / floor; stop floor 0.4%	Every approval dying at a floor another layer never saw
Crypto net-edge gate	target must clear +0.01% over 0.62% round trip	Fee-negative setups
Confidence floor with additive bumps	base 70 · crypto 35 · STMS 75 · earnings +5/+15 · probation +10 · crowding +3/+6/+9 · recovery bump while a halted book earns back	Marginal signals in hostile context
Regime posture may tighten stops, never loosen	multiplier 0.5 - 1.0	A market read quietly increasing risk
Holdings cache shorter than the monitor tick	45s vs 60s	A stale holdings set phantom-closing real positions

THE KILL-SWITCH ORDERING IS NOW A GUARDED INVARIANT

The four-day outage was an ordering bug: the per-book kill-switch block ASSIGNED recovery_bump 150 lines below the confidence sum that READ it, so every directional signal crashed inside the gate. The block now runs above the bar, the fix was reproduced live before and after, and an AST-based guard suite pins the order so it cannot silently regress.

TWO GATES FROM THE FIRST EDITION STILL STAND AS WRITTEN

The per-user daily loss limit still vetoes signals for EVERY book when any single user is over their limit — the last place one account's state freezes the platform. And MIN_PRICE remains 0.0 by intent while three comments still describe a live \$5 floor. Both carried forward as known, unfixed truths.

3.5 Asset-class policy and options preconditions

Unchanged in structure from the first edition: policy describes the instrument, never the trader's preference; an unregistered asset class fails closed to UNKNOWN_POLICY. Stocks carry native brackets; crypto uses resting stop_limit (gtc) orders at the venue (and, flag-gated, resting take-profits); options and forex have no stop type. The options preconditions still hold: market open, wheel_auto_execute on (defaults off), account not halted (fails

safe), Alpaca approval level sufficient. Wheel collateral and duration by posture: growth 25% / 1 CSP / 21 DTE · balanced 40% / 2 / 35 · income 50% / 3 / 45 · velocity 25% / 2 / 10.

4 The formulas

Every number the engine computes to decide a size, a price, a target or a probability. Written so they can be checked by hand.

4.1 Trade geometry

```
reward_to_risk      RR = target_pct / stop_pct
break-even win rate needed_win_rate = 100 / (1 + RR)      1:2 -> 33.3%   1:0.35 -> 74%
```

Verdict bands unchanged: $RR \geq 2.0$ healthy, ≥ 1.5 workable, ≥ 1.0 thin, below 1.0 upside-down. **The floor for acceptance is now the BOOK'S OWN** (`bot_settings.min_reward_risk`, seed 1.5, clamp 0.3–3.0), and the sizer reads the same row the harmoniser shaped the stop against — with one 2-decimal rounding step of grace (0.011) so cent-rounded prices cannot shave 0.40 to 0.39 and self-reject. The rescale drift detector is unchanged (reports at 0.02 ratio drift, never blocks). The harmoniser still runs last:

```
if target_pct / stop_pct < min_reward_risk(book):
    stop_pct = max( target_pct / max(min_reward_risk, 0.1), 0.004 )
```

WHY PER-BOOK MATTERS — THE SEPTEMBER REJECTION LOOP

Primary's Bot Tuning floor was 0.4; the ambient row said 0.5. The harmoniser shaped every stop to land exactly AT 0.4, then sizing enforced 0.5 — so every equity approval on that book died at execution, 100%, forever, with a perfectly "honest" message. Same disease, fourth instance: a correct per-book value existed and a call site read the ambient one. The rule that came out of it: every settings read in a per-book path takes `user_id`, and a guard suite proves two books get two verdicts.

4.2 Position sizing

```
stop_distance = |entry_price - stop_price|
eff_risk      = clamp(risk_pct, 0.0005, 0.25)      # slider; fallback 0.01
risk_usd     = equity * eff_risk
risk_qty     = risk_usd / stop_distance

cap_pct      = book's max_position_pct if set, else tiered by equity (3.1)
notional_cap = equity * cap_pct; if buying_power > 0: min(cap, buying_power)

qty = risk_qty
if qty * entry_price > notional_cap: qty = notional_cap / entry_price
if asset is not crypto:             qty = floor(qty)    # reject if < 1 share

actual_risk = qty * stop_distance
reward_risk = |target - entry| / stop_distance    # must clear the BOOK's floor
```

The sizer still rejects outright on non-positive equity, non-positive prices, a zero stop distance, or a share count that rounds below one — and now names the book whose floor it enforced.

4.3 Volatility and ATR-derived geometry

Unchanged from the first edition — true range and $ATR(14)$; overextension $|close - SMA20| / ATR14 > 4.0$ vetoes; scalp geometry `stop = max(0.004, 0.8×atr_pct)`, `target = max(0.006, 1.0×atr_pct)`; the realistic cap `min(1.5×atr_pct, learned_achieved_move)` so targets stay inside what the name has historically done. Realised volatility for option pricing is Yang–Zhang (annualised, clamped 0.15–2.00), falling back to close-to-close.

4.4 Capitalisation tiers

TIER	MARKET CAP	STOP ×	TARGET ×	SCALP ALLOWED
mega	≥ \$200B	0.80	0.65	yes
large	≥ \$10B	0.90	0.75	yes
mid	≥ \$2B	1.00	0.80	yes
small	≥ \$300M	1.25	0.80	yes
micro	< \$300M	1.60	0.80	no
etf	—	0.85	0.70	yes
unknown	data miss	1.00	0.85	no

4.5 The confidence score (TCS)

```
technical = (pattern_score / 100) * 300
options   = 250                                if 30 <= iv_rank <= 60
          = 250 * (iv_rank / 30)                if iv_rank < 30
          = 250 * max(0, (100-iv_rank)/40)       if iv_rank > 60
          = 100                                if iv_rank unknown
fundamental = 200 if catalyst_today else 80
rr          = clamp(rr_ratio * 60, 0, 150)      (110 when unknown)
market      = 50 + 30*(spy_trending_up) + 20*(confluence >= 30)

TCS = clamp( round( (technical + options + fundamental + rr + market) / 10 ), 0, 100 )
```

Computed in exactly one place, consumed everywhere as a threshold. The structural reward:risk feeding it is measured from real swing structure (risk to the swing low floored at 0.75 ATR; reward to the swing high or 2 ATR, whichever is larger).

4.6 – 4.10 Expectancy, optimal f, HRP, significance, CUSUM

All five are unchanged from the first edition and remain the measurement vocabulary: R-multiples and expectancy (with the blown count — R worse than -1 — as the execution-quality tell), Ralph Vince's optimal f with the drawdown-constrained companion, hierarchical risk parity for allocation guidance (trustworthy only at $N(N+1)/2$ observations, return-blind, paired with the expectancy screen), probabilistic/deflated Sharpe with the implied-precision solver, and the CUSUM structural-break test (5% one-sided critical value; measured false-positive 3.5% on random walks).

4.11 Trading costs

```
half_spread = ((ask - bid)/2) / mid
round_trip  = 2*(fee_bps/10000) + 2*half_spread + 2*(slippage_bps/10000)
viable      = target_pct >= round_trip * safety_multiple      # default 2.0
```

Model constants unchanged (5 bps slippage per side, \$0 stock commission, 26 bps crypto commission → 0.62% modelled crypto round trip). **New: the model is now checked against the venue.** A real-cost observer measures each coin's actual round trip at decision time — fee 0.52% plus the live spread — and logs it beside the model, observe-only. ETH runs ~0.53–0.56%, SOL ~0.62%, DOT ~0.72–0.75%, and LTC ~1.10–1.13% (spread-dominated) — the flat model flatters the illiquid coins, and acting on the per-coin measurement is a proposal, not a patch.

4.12 Option pricing

Black–Scholes, European, no dividend term, risk-free rate hardcoded at 4.3% — unchanged, formulas as in the first edition (d1/d2, put-call, gamma, vega per IV point, theta per calendar day). Implied volatility, where a live premium is available, is solved by bisection on $[0.01, 5.0]$ to $1e-5$.

THE WHEEL'S VOLATILITY EDGE IS NO LONGER ZERO BY CONSTRUCTION — WHEN QUOTES FLOW

The first edition noted implied was set EQUAL to realised, making the modelled premium ratio identically 1.0. The engine can now back implied volatility out of a REAL market premium (vol_edge.implied_vol_from_price), and production logs show distinct implied-vs-realised reads (e.g. TFC 21.16% implied vs 22.37% realised → CHEAP, veto; VTR 37.0% vs 22.84% → RICH, sellable). The equal-to-realised fallback remains wherever no live quote is available, and the go/no-go still runs on the modelled number in that case — the residual honesty gap is stated in 7.2.

Variance premium bands unchanged: $RICH \geq 1.20 \cdot FAIR > 1.00 \cdot CHEAP$ otherwise.

4.13 The adoption clock (new)

When the reconciler adopts a position the executor missed, the row's entry_at is no longer just now(): it becomes the **earliest fill of the order that built the holding**, taken from the broker's own activity records — but only from evidence inside the QA lookback window (72h).

```
evidence window = [now - TREZO_QA_LOOKBACK_H, now]      # 72h default
entry_at = earliest opening fill of the resolved order    # else stays now()
REFUSE any timestamp outside [window_start, now]        # asserted, not argued
backdated_min = (now - entry_at) in minutes              # <= lookback by bound
```

House rule: a WRONG entry_at is worse than a late one — a late one only makes a position look young, a wrong one silently moves an exit. So every unresolvable case (no order, several orders, a failed read, a fill without a timestamp) keeps now() and says why in the row. The backdate shield in 3.3 exists so an honestly-backdated row is not market-sold by a time rule on its first tick.

5 The strategies

Nine lanes. Above them sit the same three layers as before: a regime classifier (SPY into six states), an adaptive posture layer that can only ever tighten, and a per-stock selector that drops any strategy with a negative track record on that specific ticker.

5.1 – 5.3 The stock lanes

STMS (small-cap momentum), **ORB** (opening-range breakout on eight liquid names) and **Extended** (multi-day swing, four continuation setups, the one lane that holds overnight) are unchanged in trigger, exit and guard-rail from the first edition — including Extended's profit ladder (+4% lock breakeven, +7% lock +3%, +10% lock +6%), its 7-day time stop, its FOMC entry blackout, and ORB's known prose-vs-constants disagreement (the constants run 8:30 ET pre-market as "best"; the constants are what runs). What changed around them: the widened stock pockets (§1.2) give all three lanes more chairs, and their refusals at full pockets are now audible instead of silent.

5.4 Crypto — four modes, one scanner, now a real venue

Long only, 24/7, four mutually exclusive modes in priority order (SWING, HODL, DCA, SCALP), every entry through the fee-edge gate first — all unchanged, including the profit ladders (SWING ratchets at +0.8/1.0/1.8/3.0/5.0/10%) and the documented SCALP geometry distortion ($0.6\times$ stop / $0.5\times$ target; reported, deliberately not blocked). What changed is the plumbing under it (§1.5): orders go to Alpaca paper crypto; stops REST at the venue and ratchet; the stop's quantity is the venue's own position quantity, never the ledger's rounding.

TWO LIVE DEFECTS THIS VENUE EXPOSED

(1) A new order on a coin whose stop is resting trips Alpaca's wash-trade check — HTTP 403, "use complex orders" — so adds and re-entries on stop-protected coins currently die at the broker. Cancel-replace around entries, skip adds on protected coins, or adopt complex-order semantics: an open design decision. (2) With TREZO_CRYPTOREEVAL off, losing coins are never re-scored, so a pocket full of losers stays full — the September capacity lock. The re-score ships built and dormant; enabling it is a one-line env change.

5.5 The Dividend Wheel

Unchanged in structure: CSP at strike $\approx$ spot $\times$ 0.95 targeting 30 DTE (decay override projects the strike DOWN, never up); covered calls at $\max(\text{spot} \times 1.05, \text{cost_basis} \times 1.01)$ with a 7.5% cushion across an ex-dividend date; recovery CCs for lots $\geq 15\%$ underwater under three strict conditions; the Wheel Advisor still subtracts only (defer or shrink), still fails open, still runs the ■ checks (5–45 DTE, ± 2 -day earnings blackout, ex-date guard, collateral honesty). GROWTH-tier names can never wear a call. The advisor's shrink branch remains mathematically unreachable while the CSP path requests one contract — a known, held-for-redesign item.

5.6 – 5.7 Options structures and cycle-aware biases

The eight defined-risk builders (35 DTE, fixed strike offsets, net position Greeks) and the two selector-only biases (IV crush short into earnings windows; dividend capture long around ex-dates, both deliberately bypassing the earnings bump) are unchanged.

5.8 Forex — declared, still venue-less, now with real candles

The ten Kraken pairs are scored on REAL 4-hour candles now (Twelve Data primary, Kraken public data fallback) rather than a synthetic feed — but there is still no venue that can hold an FX position, so every signal still dies at the gate. The lane remains overhead plus veto noise that hides real rejections; give it a venue or turn it off. TREZO_FOREX_ENABLED defaults off.

5.9 The strategy library

Unchanged: fifteen canonical strategy cards and a regime playbook as a read-only knowledge base; six map to live modules; the other nine are described and never traded, as philosophy rather than backlog. The agents also read a document library on disk (TREZO_DOC_DIR, 40MB per-file cap) — books and notes that inform research context, not rules.

6 The Dividends (Long-Term) lane

Unchanged in every formula since the first edition — restated here in brief so this revision stands alone; eb271bb Part 6 remains the full treatment.

The lane splits its income-pocket capital three ways — ladder L (50–90%), wheel W (0–40%), buffer B (3–20%) — with HARD guard rails that raise rather than clamp. Ladder names = clamp(floor(L/1000), 1, 15); CSP blocks = floor(W / \$3,500); expected_TR = w_ladder×(5.3%+4.5%) + w_wheel×8.0% (still labelled UNPROVEN) + w_buffer×3.75%. The per-name state machine (FRACTIONAL → LOT_READY / LOT_HELD → CASH_SECURED → ASSIGNED) still decides what is permitted, the five lane rules still bind (two-state names; graduation at 100 shares; the ex-date guard; no covered calls on GROWTH names; hard collateral reservation), and the entry screen still runs the company rule (yield ≥ 1.5%, payout ≤ 70%, unbroken raise streak with the repaired-cut doctrine) and the fund rule (distribution funded by total return, \$2M/day volume floor, no reverse split in 24 months). Income mode still cannot draw principal: draw = max(0, min(actual distributions, 90% of trailing 12-mo total return)). The corporate-actions dividend feed (broker-sourced, ex-dates and special flags) remains the live data source.

7 Limitations

What the engine cannot do, cannot see, or does not really do despite appearing to. Four categories, unchanged: a data gap is closed by buying or building a feed; a modelling approximation by measurement; inert code by deleting or wiring it; a deliberate deferral is a decision, listed so nobody re-discovers it as a surprise.

7.1 Data-source gaps (updated)

WHAT IS MISSING	WHAT DEPENDS ON IT
No option Greeks or IV from any feed — but IV is now solvable from live premiums where quotes flow (4.12)	Wheel strikes and delta reasoning. Delta is computed, never vendor-observed. Historical option daily bars reach back only to Feb 2024 (probed); nothing wired reaches earlier.
No point-in-time fundamentals	Every backtest screening on fundamentals carries lookahead bias; the payout screen runs historically only at today's snapshot.
Corporate-actions history begins 2016	The literal ten-year raise streak tightens on its own as history accumulates.
No real macro source	VIX and treasury reads fall through to ETF proxies in scoring; the Treasury.gov daily curve is wired for measurement work.
Top-of-book artifacts from the market-data feed	Hundreds of vetoes a day are data quality, not signal quality — the 2% divergence guard bounds it.
No true slippage measurement for MODELLED fills	Only crypto now has real fills; stock paper fills are Alpaca-simulated; the modelled engine's 5 bps remains a model.

7.2 Modelling approximations that remain

APPROXIMATION	WHAT RESTS ON IT
BS everywhere, European, rate hardcoded 4.3%	Every wheel go/no-go where no live quote refines it.
Dividend-lane calibration constants are placeholders	Expected return, monthly income, the target readout. A measurement program (v2, on probed Alpaca+Treasury rails) exists to replace them; first pilot ran; full run pending.
All trading windows are UTC approximations	Half the year each window is an hour wider than intended — now including the 19:45Z force-exit and the three scheduled market-report crons. Accepted, stated, and slated for America/New_York math in any rebuild.
Dividend distributions modelled quarterly	Understates weekly/monthly funds; live ex-date fetch written, untested against production payloads.
The crypto ladder replay knows entry, peak and exit — not the path	Its rungs remain optimistic by an unknown amount.
HRP covariance still cannot honestly be computed	Reported with a trustworthiness flag, never as fact.

7.3 Known-inert and known-wrong code (updated)

ITEM	STATE
Staleness tiering	FIXED — rescaled to the 0–100 score (3.2).
Lane rules 3 and 4 (ex-date, growth-call)	Fixed in the first edition's window; still live.
Watchdog roster gap (23/29)	FIXED — 30/30, asserted by guard (Part 2).
ORB/Extended emitting 0–1000 confidence	Still present: their internal floor cannot bind; signals clear the 0–100 gate unconditionally. Same rescale-miss family, two modules.
The scheduler's intended start path never executed	Still true (fallback boots the engine; one boot log signal lies).
user_support	Still a registered stub.
Dead modules / never-called public functions	Still present (~734 lines / 43 functions at last count); groundwork vs abandonment remains indistinguishable without this note.
Phantom P&L; residue	NEW: ~\$5.8k of DOT closes that never happened at the broker still sit in realized counters; QA now PROVES phantom closes (qa_phantom_close_proven) but the counter cleanup has not shipped.
Per-user daily loss freeze / MIN_PRICE comments	Carried forward, unfixed (3.4 callout).

7.4 Deliberate deferrals (updated)

Carried forward: ORB credit spreads; intraday penny patterns (need an intraday feed); acting on the variance-premium measurement (behaviour change → proposal); nine of fifteen library strategies; live brokerage execution (scaffolded behind a go-live checklist); crypto short side; broker-routed partial closes on bracket-attached positions; trusting the dividend-lane ranking; the twelve-month verdict gate. New deferrals: enabling

TREZO_CRYPTOREEVAL (Mike's call — it can exit losing coins); the wash-trade executor redesign (5.4); counting capacity into the approval gate itself so the same six names stop being approved every five minutes for the executor to refuse.

7.5 What the record says now

THE PROFITABLE LANE IS NO LONGER THE LEAST VALIDATED ONE

The first edition noted the lane carrying the platform (crypto) was the one whose fills its own engine simulated. That inversion is gone: crypto fills at the real paper venue now, with real rejections. What replaced it as the honest caveat: recent realized P&L; has been poor (55 closes for $-\$1,267$ on 2 September, profit factor 0.03 that day) while the books sit at capacity — the platform currently proves its SAFETY rails far better than its EDGE. The measurement program, not this reference, is where the edge question gets answered.

The reading rule from the first edition stands verbatim: none of the failures in 7.3 ever produced an error. The only defence is to periodically ask each guard when it last actually fired — a guard that has never fired in production is either unnecessary or broken, and it is worth knowing which. The nets in 1.4 exist to ask that question continuously.

8 The sixteen-day record

Everything that changed between eb271bb (23 August) and 4163078 (8 September), in order, each with the scar it left as a rule. This part is why the second revision exists.

DATE	WHAT HAPPENED	WHAT IT LEFT BEHIND
8/27	52-item audit; same-morning clear-out claimed 22 fixes; adversarial second pass found 8 hold, 7 partial, 6 no-bind, 8 new defects	Verify a fix by reading the RESULTING file, never the diff. "Fixed" is a claim until a guard proves it.
8/27–31	THE OUTAGE: kill-switch ordering bug crashed every directional signal inside the gate; router swallowed it; zero approvals for four trading days	Nets 1 & 2 (\$1.4); ordering pinned by AST guard; the house rule that a quiet log is not a healthy log — count approves.
8/28	After-close batch: the alert channel exists and is PROVEN (send_test, then a real alert delivered same night); max-notional passthrough; per-agent tick ceilings; USB backup purge guards	TREZO_ALERT_WEBHOOK; the platform can finally reach a person.
8/29	The phantom-close loop: one failed positions read returned [], monitor closed the whole 75k book at modelled prices, reconciler re-adopted, ~\$5.8k booked that never happened; crypto stop 403s from 8dp-vs-9dp rounding	Broker truth: None ≠ []; strict reads; never cache a failure; venue spelling (DOTUSD vs DOT/USD); venue's own quantity; parse the 403's 'available' and retry once. DOT's stop finally rested at the broker 21:51Z.
8/29	First deploy of that fix ROLLED ITSELF BACK (pytest fixtures in a bare-runner gate); second deploy went green and changed nothing (wrong endpoint spelling, swallowed)	pytest-green ≠ gate-green; a green deploy is not a working fix — watch the live log for the symptom to STOP.
8/31	Outage root-caused and fixed live (gate moved above the bar); 10 approvals and 5 fills within 15 minutes of the deploy; both nets shipped	Incident write-up on disk; test the message PATH, not the policy function.
9/2	R:R floor split closed (per-book floors, threaded through all three call sites); posture split widened every stock pool (Mike); losing-coin 24/7 defense shipped (re-score flag dormant); reevaluator first-hour fix; approval-starvation net FIRED correctly in production and delivered	Built-but-not-bound named as the house failure mode, fourth instance; grep any bare settings read in a per-book path.
9/2–3	Twin-session waves: trade QA inspector (reconciles broker receipts vs ledger; proves phantom closes), book-what-arrived crypto fills, adoption keeps the broker's clock (entry receipts, 4.13), per-book stacking dedup, audible capacity refusals	Every deliberate refusal writes an activity row naming book, pocket and numbers.
9/4–7	EXECUTION STARVATION alerts all weekend: 18–21 approvals per window, zero fills — all deliberate capacity refusals the flow alarm could not see; plus 2 real wash-trade 403 kills on 9/4	CAPACITY LOCK alert shape (warn, actionable, dedup while it persists) — on main, awaiting deploy. Wash-trade collision promoted to open design decision.
9/7	Gate de-decay: the adoption-clock suite's fixtures aged out of their own 72h evidence window and its shield tests failed after 15:45 ET daily — a poisoned gate that would roll back ANY deploy	Gate suites must not carry an expiry date: frozen clocks, no wall-time assertions.

Companion documents to this revision: the **Rebuild Blueprint** (8 September — prerequisites, env inventory, migrations, bring-up order, keys by name) and the **Measurement Program v2** (probed Alpaca + Treasury rails; replaces the dividend-lane placeholders with measurements). The full source travels as trezo-platform-main-4163078.zip; the repo history is clean of secrets.

A Constants quick reference

Every number that changes behaviour, in one place, at its deployed value. Changes from eb271bb are marked ◆.

A.1 Risk and capital

CONSTANT	VALUE	EFFECT
Risk per trade	slider (clamp 0.05% - 25%) · fallback 1% ◆	Dollars risked per position
Reward:risk floor	PER BOOK ◆ · seed 1.5 · clamp 0.3 - 3.0 · tolerance 0.011 ◆	Below the book's floor the trade is rejected (books today: 0.4 / 0.5 / 0.5)
Absolute stop floor	0.4%	The harmoniser will not tighten past this
Target floor	0.6% (TREZO_TARGET_MIN_PCT)	No sub-fee targets
Max open positions per book	14 ◆ (was 3)	New-ticker cap
Total deployment ceiling	1.25 × equity	Leverage bound
Per-position notional cap	15 / 30 / 25 / 15%	By equity band, smallest to largest
Crypto per-coin cap	10% of equity	Summed across all open rows
Crypto accumulation cooldown	18 hours	Spacing between adds
Approval TTL	2 hours · PER BOOK ◆	Duplicate-approval guard
Hopeful-bucket cap	3% of options capital	Low-probability structures
Small-account pocket stretch	1.75×, cap 60%	Below \$25,000 equity

A.2 Kill switches and nets

TRIGGER	THRESHOLD
Daily realised drawdown	3% of day-start equity
Weekly realised drawdown	6% of week-start equity
Consecutive losses, per book	3
Broker rejects	3 within 60 minutes
Slippage breaches	3 fills worse than 75 bps
Per-coin daily loss bench	10% of coin budget (advisory)
◆ Approval starvation (per lane)	window ≥ 20 min AND ≥ 15 signals AND 0 approvals (lane's market hours; crypto 24/7)
◆ Execution starvation (per lane)	≥ 3 approvals AND 0 fills; kills and deliberate refusals accounted separately
◆ Capacity lock (per lane)	all approvals refused on purpose → ONE warn while it persists (on main, awaiting deploy)

TRIGGER	THRESHOLD
◆ Alert quiet times	urgent 30 min · warn 4 h · info 24 h, per key

A.3 Filters and windows

FILTER	VALUE
Max ATR stretch from SMA20	4.0
Max bid/ask spread	0.5%
Global average-volume floor	250,000 (per-strategy: STMS 100k · ORB / Extended / Wheel 500k)
Tape freshness	20 minutes
Stale-book divergence	2%
Intraday max hold	90 minutes
Stagnation stop	75 minutes below 0.25R
End-of-day force exit	19:45 UTC (EDT-anchored; drifts under EST)
Swing max hold	7 days
◆ Signal staleness (rescaled, now fires)	$TCS \geq 70 \rightarrow 60 \text{ s} \cdot 50-69 \rightarrow 180 \text{ s} \cdot \text{else } 300 \text{ s}$
Confidence floor	70 base · 35 crypto · 75 STMS
Max confidence bump	15
Stop multiplier bounds	0.5 – 1.0 (tighten only)
Book-scope cache TTL	45 seconds (monitor ticks at 60)
◆ Entry-receipt evidence window	72 h (TREZO_QA_LOOKBACK_H); timestamps outside [window_start, now] refused

A.4 Options and the wheel

CONSTANT	VALUE
Target DTE	30
CSP / CC out-of-the-money offset	5% (7.5% across an ex-date)
Advisor DTE window	5 – 45
Earnings blackout	±2 days
Posture collateral caps	growth 25% · balanced 40% · income 50% · velocity 25%
Posture open-CSP caps	1 / 2 / 3 / 2
Posture DTE caps	21 / 35 / 45 / 10

CONSTANT	VALUE
Decay override trigger	-2% monthly drift, 25% safety margin
Recovery distress threshold	15% below basis
Recovery minimum credit	$\max(\$10, 0.2\% \text{ of lot value})$
Minimum DTE for any option	7
Risk-free rate	4.3%
Implied-volatility clamp	15% - 200% (solver 1% - 500%)
Variance-premium bands	$\text{rich} \geq 1.20 \cdot \text{fair} > 1.00 \cdot \text{cheap}$ otherwise $\blacklozenge$ implied from live premium when quotes flow

A.5 – A.6 The dividend lane and screen

Unchanged from eb271bb, every value: ladder/wheel/buffer bounds 50–90% / 0–40% / 3–20%; wheel delta cap 0.15–0.40; target-return bounds 5–25%; minimum lane capital \$500; max 15 ladder names at \$1,000 each; CSP block cost \$3,500; per-name cap 20% tightening to 10% at twelve names; assumed ladder yield/growth 5.3%/4.5%; assumed wheel total return 8.0% (unproven); buffer 3.75%; income-mode draw ceiling 90% of trailing total return. Screen: yield $\geq 1.5\%$; growth/high-yield split at 4.0%; payout $\leq 70\%$; 10-year raise streak (window-limited) with the repaired-cut doctrine (recovered + rising + 3 healed years); fund liquidity floor \$2M/day; reverse-split lookback 24 months; special-dividend detection 3× median; sector cap 2 names.

A.7 Cost model

CONSTANT	VALUE
Slippage, per side (model)	5 bps
Stock commission	\$0
Crypto commission (model)	26 bps
Crypto round trip (model)	0.62%
$\blacklozenge$ Crypto round trip (measured, live)	$\text{fee } 0.52\% + \text{live spread} - \text{ETH } \sim 0.54\% \cdot \text{SOL } \sim 0.62\% \cdot \text{DOT } \sim 0.73\% \cdot \text{LTC } \sim 1.11\%$
Crypto minimum net edge	+0.01% over round trip
Viability safety multiple	$2.0 \times \text{round-trip cost}$

A CLOSING NOTE ON NUMBERS, SECOND EDITION

Most of the thresholds above are still scars, and this revision added new ones: the 0.011 tolerance exists because cent-rounding shaved 0.40 to 0.39 and self-rejected a healthy book; the per-book approval TTL exists because one book's approval silenced every other book's same idea; the capacity-lock alert exists because an urgent alarm that cries wolf at a correctly-full book trains its reader to ignore the channel — which is how the LAST outage survived four days. A number without its reason is a number somebody will optimise later. That is why this document states both.

Prepared by Nova for Mike · commit 4163078 · 8 September 2026 · second revision, continuing eb271bb of 23 August 2026. Every threshold was read from the deployed source rather than from design notes; Part 8's claims are drawn from the engine's own activity record, the incident write-ups on disk, and alerts that really fired.